Integration Test Report

StreetRace Manager

1 Introduction

This document records integration test cases across modules, including expected and actual results, detected issues, and fixes.

2 Why Integration Tests Are Needed

Integration tests are required because modules interact with each other. Even if a module works independently, failures may occur when data flows between modules.

Examples:

- Registration may work, but Race Management must reject unregistered drivers.
- Results may work, but prize updates may fail due to Inventory issues.
- Mission Planning may work, but CLI input may block full crew assignment.

3 Test Case Matrix

ID	**Scenario Being Tested**	**Modules Involved**	**Expected Result**	**Actual**	**Erro**
IT-01	CLI mission assignment accepts yes for multiple crew	CLI, Mission Planning, Registration, Crew Management	All entered crew IDs are attached to mission	PASS	CLI (yes/
IT-02	Role assignment without registration	Registration, Crew Management	Must fail	PASS	None
IT-03	Registration followed by role assignment	Registration, Crew Management	Role assigned	PASS	None
IT-04	Registered crew details retrieval	Registration	Consistent lookup	PASS	None
IT-05	Unregistered ID attempts to enter race	Crew Management, Race Management	Rejected	PASS	None
IT-06	Non-driver crew attempts to enter race	Registration, Crew Management, Race Management	Rejected	PASS	None

ID	Scenario Being Tested	Modules Involved	Expected Result	Actual	Erro
IT-07	Driver registration to race participation	Registration, Crew Management, Race Management, Inventory	Driver + vehicle recorded	PASS	None
IT-08	Race start readiness transition	Registration, Crew Management, Race Management, Inventory	Valid start only	PASS	None
IT-09	Duplicate driver entry into same race	Registration, Crew Management, Race Management	Reject duplicate	PASS	None
IT-10	Race result deducts prize from inventory cash	Registration, Crew Management, Race Management, Results, Inventory	Cash decreases	PASS	None
IT-11	Race result fails on insufficient funds	Results, Inventory	Error raised	PASS	None
IT-12	Race win updates crew ranking counters	Registration, Crew Management, Race Management, Results	Wins updated	PASS	None
IT-13	Assigning unregistered crew to mission	Registration, Mission Planning	Rejected	PASS	None
IT-14	Rescue mission with required roles	Registration, Crew Management, Mission Planning	Valid start	PASS	None
IT-15	Heist mission missing roles	Registration, Crew Management, Mission Planning	Blocked	PASS	None
IT-16	Surveillance mission requires strategist and spotter	Registration, Crew Management, Mission Planning	Partial fail then pass	PASS	None
IT-17	CLI add driver and vehicle to race	CLI, Registration, Crew Management, Race Management, Inventory	Driver + vehicle added	PASS	None
IT-18	CLI record result updates result/reputation/cash	CLI, Race Management, Results, Inventory, Reputation	Updates all systems	PASS	None
IT-19	Repair operation deducts cash	Inventory, Maintenance	Cash reduced	PASS	None
IT-20	Vehicle condition improves after repair	Inventory, Maintenance	Condition improved	PASS	None
IT-21	Damaged vehicle cannot be added to race	Inventory, Race Management	Rejected	PASS	None

ID	Scenario Being Tested	Modules Involved	Expected Result	Actual	Erro
IT-22	Reputation initialization	Registration, Reputation	Default created	PASS	None
IT-23	Mission success increases reputation	Registration, Reputation	Score increases	PASS	None
IT-24	Mission failure decreases reputation	Registration, Reputation	Score decreases	PASS	None
IT-25	Reputation level progression	Registration, Reputation	Correct level	PASS	Basel
IT-26	End-to-end racing workflow	Registration, Crew Management, Inventory, Race Management, Results, Reputation	Full consistency	PASS	None
IT-27	End-to-end mission workflow	Registration, Crew Management, Mission Planning	Completed lifecycle	PASS	None
IT-28	Vehicle damage-repair-rerace workflow	Inventory, Maintenance, Race Management, Crew Management, Registration	Correct transitions	PASS	None
IT-29	Crew skill helper APIs	Registration, Crew Management	Works correctly	PASS	None
IT-31	Maintenance helper APIs and cost tracking	Inventory, Maintenance	Accurate totals	PASS	None
IT-32	Mission metadata exposes required roles	Registration, Crew Management, Mission Planning	Correct roles	PASS	None
IT-33	Reputation listing ordered by score	Registration, Reputation	Sorted output	PASS	None
IT-34	Listing of recorded race results	Results, Inventory	Correct output	PASS	None
IT-35	Inventory cash helpers and listings	Inventory, Registration, Crew Management, Race Management, Mission Planning	Correct updates	PASS	None
IT-36	needs_maintenance helper reflects damage	Inventory, Maintenance	Accurate status	PASS	None
IT-37	Leaderboard ordering by race wins	Results, Inventory	Sorted by wins	PASS	None

4 Error Log (Recorded Before Correction)

4.1 Error E1 - CLI mission assignment stops early

- **Scenario:** Assigning crew in mission flow with answer `yes`
- **Expected:** Additional crew assignment continues
- **Actual:** Loop stopped because code checked only exact `y`
- **Evidence:** Failing integration test `test_create_mission_allows_yes_for_multiple_crew_assig` with `AssertionError: 0 != 3`
- **Fix Applied:** CLI now accepts both `y/yes` and `n/no`
- **Re-test Result:** PASS

4.2 Error E2 - Repair integration failed without parts in test setup

- **Scenario:** Repairing damaged vehicle in integration tests
- **Expected:** Repair succeeds when logic is correct
- **Actual:** Failed due to missing `standard_parts` inventory
- **Fix Applied:** Added required parts (`standard_parts`, `advanced_parts`, `oil`, `filters`)
- **Re-test Result:** PASS

4.3 Error E3 - Reputation level baseline mismatch

- **Scenario:** Initial reputation level assertion
- **Expected:** Rookie
- **Actual:** Default score 50 maps to Professional
- **Fix Applied:** Updated test expectation
- **Re-test Result:** PASS

4.4 Additional Detected Issues (H-series)

- **H1 - Skill lookup KeyError in Crew Management**
  - Tests: `test_set_and_get_crew_skills_and_levels` (IT-29)
  - Defect: `get_skill_level()` raises KeyError instead of returning 0
- **H2 - Tools missing from inventory snapshot**
  - Tests: `test_add_tools_and_list_inventory` (IT-30)

  - Defect: `list_inventory()` omits tools section and shows incorrect quantities

- **H3 - Maintenance total cost miscalculated**
  - Tests: `test_perform_standard_maintenance_and_total_cost`, `test_get_repair_cost_lool` (IT-31)
  - Defect: Total cost not accumulated properly and incorrect cost for unknown repairs

- **H4 - Cash helpers and listings inconsistent**
  - Tests: `test_add_cash_and_deduct_cash_round_trip`, `test_list_races_and_list_missions` (IT-35)
  - Defect: Negative values allowed; listings return only latest entry

- **H5 - Leaderboard not ordered by wins**
  - Tests: `test_leaderboard_orders_by_wins` (IT-37)
  - Defect: Returned in insertion order instead of sorted order

4.5 Final Verification

- Command used: `python -m unittest -v test_integration.py`
- Result: `Ran 39 tests ... OK`
- Conclusion: All module interactions are working correctly. Previously detected integration issues have been resolved.

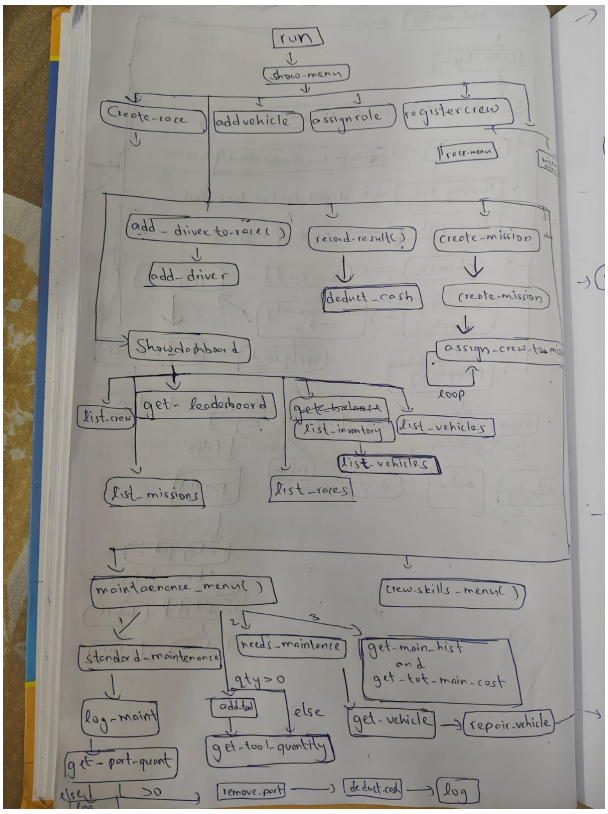

Figure 1: Q11 Image

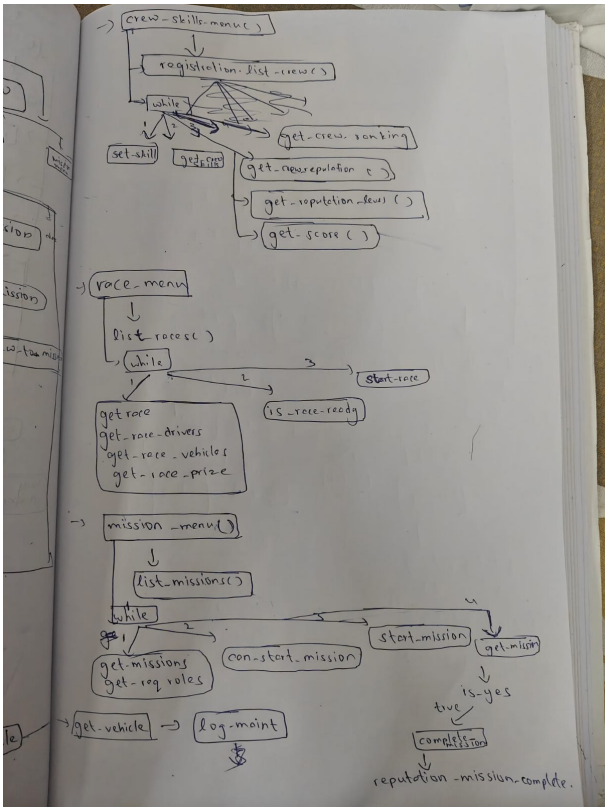

Figure 2: Q12 Image